Xuan Tan

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RESEARCH INTERESTS

Dexterous hands for everyday manipulation: grasp planning, contact-aware control, force sensing

EDUCATION

University of Colorado Boulder **December 2025**

B.S. in Computer Science | GPA: 3.8 / 4.0

RESEARCH PAPERS

Grasp Planning and Contact Control for the Inspire RH56 Hand Family

Xuan Tan, William Xie, and Nikolaus Correll.

Under review at ICRA 2027. Earlier version: arXiv:2603.08988 | Project page

GOLEM: Modular Humanoid Autonomy Towards Electric Vehicle Battery Disassembly

Max Conway, William Xie, Allen Devaraj, Yutong Zhang, Niraj Pudasaini, Mateo Feit, Adam Abid, Zachary Allen, Chen Liu, **Xuan Tan**, Jensen Lavering, Jason Chen, Lyle Antieau, Anthony Von Pischke, Alessandro Roncone, Zachary Sunberg, and Nikolaus Correll.

Under review at RA-L. arXiv:2608.21550

RESEARCH EXPERIENCE

Research Assistant, Correll Lab **March 2025 – Present**

University of Colorado Boulder | Advisor: Dr. Nikolaus Correll | Full-time since January 2026

Inspire RH56 grasp planning and contact control | First author | Code

- **Characterization and control.** Calibrated the hand's built-in finger-force readings, measured command latency, and found contact-force overshoot of up to 16× the commanded limit at full speed. Designed a fast-approach, slow-contact controller that keeps overshoot near slow-speed levels.
- **Grasp planning.** Modeled the hand's coupled linkages (six actuators, twelve joints) in MuJoCo and built a planner that maps object width to finger closure and wrist pose. Analyzed where each grasp strategy can reach an object without collisions, and compared the planner with a QP-based differential-IK solver (more accurate for multi-finger grasps, but jittery).
- **Hardware evaluation.** Across 450 hardware grasps on 15 objects, planned grasps succeeded in up to 87% of trials vs. 48% when treating the hand as a parallel-jaw gripper. With real RGB-D perception, they succeeded in 34 of 70 test conditions vs. 13 for GraspGen-X, a learned grasp generator.
- **Contact sensing.** Release triggered by the hand's own finger forces completed peg-in-hole insertion in 13/20 trials vs. 2/20 with a wrist force/torque sensor. Built online grasp-quality diagnostics (grasp wrench space) that track a grasp as it degrades.
- **Open-source software.** Lead developer of rh56_controller, the lab's open-source package for controlling, simulating, and planning grasps with the hand (UR5, Unitree H1-2, ROS 2).

GOLEM humanoid autonomy | Spatial memory

- Integrated an existing FAISS-based retrieval pipeline into the spatial memory of GOLEM, a modular autonomy stack for a Unitree H1-2 humanoid that disassembles EV batteries, and ran experiments for the team's study.

SELECTED ROBOTICS PROJECTS

Supermarket Object Retrieval with TIAGo **2025**

Three-person team | Webots simulation

- Built occupancy-grid mapping, path planning, and control for autonomous supermarket navigation.
- Integrated grasp-pose generation, inverse kinematics, and motion planning for simulated shelf-to-basket object retrieval.

Sand Dunes Rover **2024**

Colorado Space Grant Consortium | Seven-person team

- Led software development and wrote Arduino motor-control and encoder firmware.
- Fused IMU and encoder measurements with an extended Kalman filter and implemented PID-based waypoint tracking.

TECHNICAL SKILLS

Programming Python, C/C++, Bash

Robotics and software MuJoCo, ROS 2, Webots, Arduino, Linux, Git, CMake, Jupyter, LaTeX

Methods Force calibration, manipulation control, kinematics, grasp and motion planning, sensor fusion, hardware experiments